ECE 332: Engineering Electromagnetics II

Lecture 7

Mark A. Martin

Portland State University

April 21, 2026

Quick Recap

- The **polarization** of a uniform plane wave describes the curve that the electric field intensity vector $\mathbf{E}$ traces in a plane perpendicular to the direction of propagation $\hat{\mathbf{z}}$ at a fixed value of z.
- The curve is the one that the **phasor $\tilde{\mathbf{E}}$ follows**.
- The phasor has the form

$$\tilde{\mathbf{E}}(z) = E_{x0} e^{-jkz} \hat{\mathbf{x}} + E_{y0} e^{-jkz} \hat{\mathbf{y}}, \tag{1}$$

  where E_{x0} and E_{y0} are complex.
- The **magnitudes** $a_x = |E_{x0}|$ and $a_y = |E_{y0}|$ and the **phase difference** δ between the coefficients E_{x0} and E_{y0} **determine the polarization**. So, the phase of E_{x0} is defined to be 0 and the phase of E_{y0} is defined to be δ. This means

$$\tilde{\mathbf{E}}(z) = (a_x \hat{\mathbf{x}} + a_y e^{j\delta} \hat{\mathbf{y}}) e^{-jkz} \tag{2}$$

  and

$$\mathbf{E}(z,t) = a_x \cos(\omega t - kz) \hat{\mathbf{x}} + a_y \cos(\omega t - kz + \delta) \hat{\mathbf{y}}. \tag{3}$$

Quick Recap (Continued)

- The electric field intensity has **magnitude**

$$|\mathbf{E}(z,t)| = (a_x^2 \cos^2(\omega t - kz) + a_y^2 \cos^2(\omega t - kz + \delta))^{1/2} \tag{4}$$

  and has the **inclination angle**

$$\psi(z,t) = \tan^{-1}\left(\frac{E_y(z,t)}{E_x(z,t)}\right) \tag{5}$$

  relative to the positive x axis, where $a_x = |E_{x0}|$ and $a_y = |E_{y0}|$.

- In general, $\mathbf{E}(z,t)$ follows an **ellipse**. But the ellipse can be a circle (**circular polarization**) or a line segment (**linear polarization**).

Quick Recap (Continued)

- **Linear Polarization**

  A uniform EM plan wave is linearly polarized when

  - $\delta = 0$ and $E_x(z,t)$ and $E_y(z,t)$ are **in phase**

$$\mathbf{E}(z,t) = (a_x\hat{\mathbf{x}} + a_y\hat{\mathbf{y}})\cos(\omega t - kz) \quad (6)$$

$$\psi(z,t) = \tan^{-1}\left(\frac{a_y}{a_x}\right) \quad (7)$$

  - $\delta = \pi$ and $E_x(z,t)$ and $E_y(z,t)$ are **out of phase**

$$\mathbf{E}(z,t) = (a_x\hat{\mathbf{x}} - a_y\hat{\mathbf{y}})\cos(\omega t - kz) \quad (8)$$

$$\psi(z,t) = \tan^{-1}\left(-\frac{a_y}{a_x}\right) \quad (9)$$

  In both cases,

$$|\mathbf{E}(z,t)| = \sqrt{a_x^2 + a_y^2}\,|\cos(\omega t - kz)|. \quad (10)$$

Quick Recap (Continued)

- If $a_y = 0$, $\psi = 0$ or π and the wave is **x-polarized**.
- If $a_x = 0$, $\psi = \pi/2$ or $-\pi/2$ and the wave is **y-polarized**.

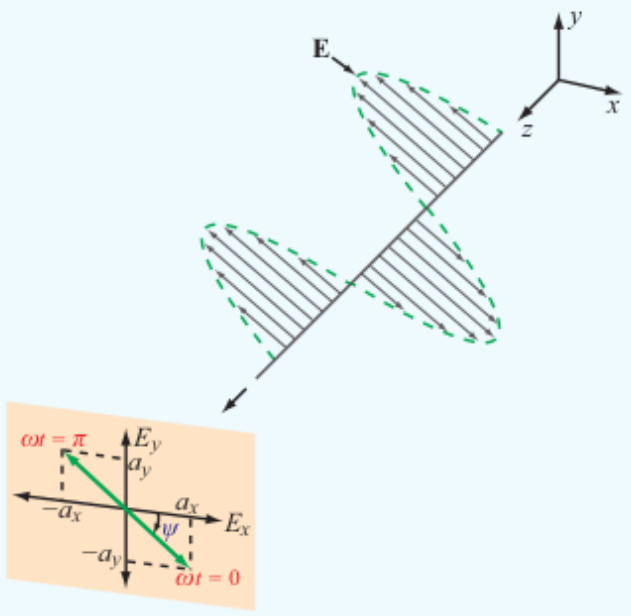


Figure 7-7 Linearly polarized wave traveling in the $+z$ direction (out of the page).

Quick Recap (Continued)

- **Circular Polarization**

  A uniform plane wave is circularly polarized when $a_x = a_y = a > 0$ and $\delta = \pm\pi/2$. In both cases, $|\mathbf{E}(z,t)| = a$.

  - **Left-Hand Circular (LHC) Polarization**

    In this case, $\delta = \pi/2$, which means

$$\tilde{\mathbf{E}}(z) = a(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz}, \tag{11}$$

$$\mathbf{E}(z,t) = a(\cos(\omega t - kz)\hat{\mathbf{x}} - \sin(\omega t - kz)\hat{\mathbf{y}}), \tag{12}$$

$$\psi(z,t) = -(\omega t - kz). \tag{13}$$

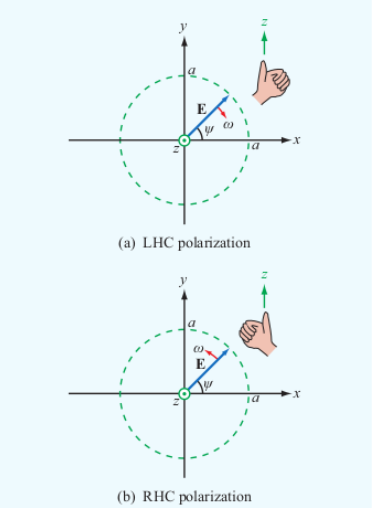


Figure 7-8 Circularly polarized plane waves propagating in the $+z$ direction (out of the page).

Quick Recap (Continued)

- **Right-Hand Circular (RHC) Polarization**

  In this case, $\delta = -\pi/2$, which means

$$\tilde{\mathbf{E}}(z) = a(\hat{\mathbf{x}} - j\hat{\mathbf{y}})e^{-jkz}, \tag{14}$$

$$\mathbf{E}(z,t) = a(\cos(\omega t - kz)\hat{\mathbf{x}} + \sin(\omega t - kz)\hat{\mathbf{y}}), \tag{15}$$

$$\psi(z,t) = \omega t - kz. \tag{16}$$

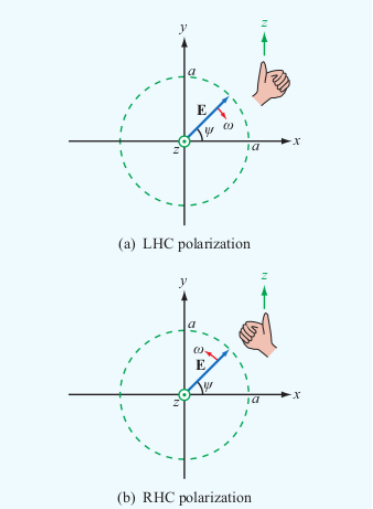


Figure 7-8 Circularly polarized plane waves propagating in the $+z$ direction (out of the page).

For both types of circular polarization, **E** rotates in the opposite sense as a function of z for fixed t as it does as a function of t for fixed z.

- We explored an example of finding **E** and **H** for a RHC polarized wave.

Quick Recap (Continued)

- **Elliptical Polarization**

In this case,

$$\tilde{\mathbf{E}}(z) = (a_x \hat{\mathbf{x}} + a_y e^{j\delta} \hat{\mathbf{y}}) e^{-jkz}, \quad (17)$$

where $a_x \neq a_y$ and $\delta \neq 0, \pm\pi/2, \pi$.

The ellipse that $\mathbf{E}(z,t)$ traces is specified by its

- **semi-major axis** a_ξ,
- **semi-minor axis** a_η,
- the **rotation angle** $\gamma \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the major axis makes with a reference direction.

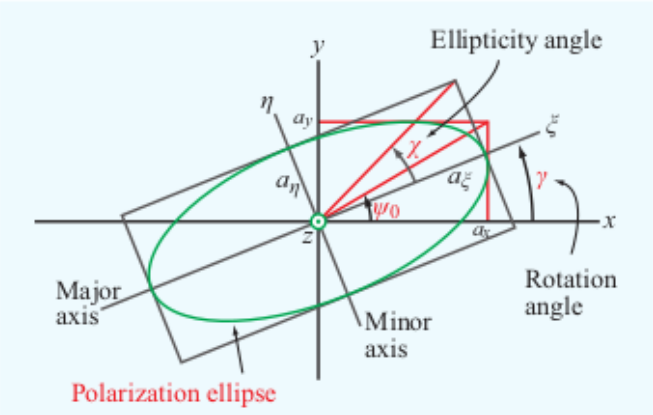


Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

Quick Recap (Continued)

- The **polarization is right- or left-handed**, following the same rules as for circular polarization.
- The **ellipticity angle** $\chi \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ specifies the **ratio between the major and minor axes** of the ellipse and the **handedness** of the polarization.
- The **axial ratio** $R = a_\xi / a_\eta$ satisfies

$$\tan \chi = \pm a_\eta / a_\xi = \pm 1/R. \quad (18)$$

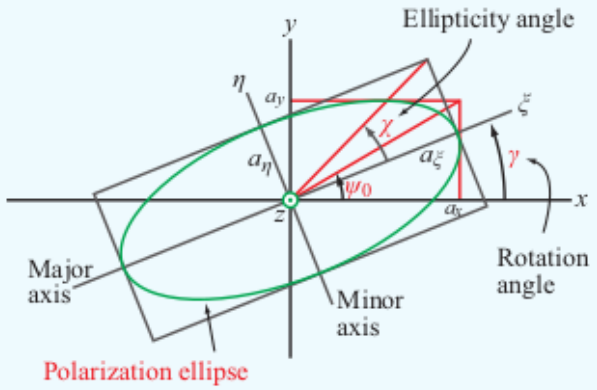


Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

Quick Recap (Continued)

- If $\tan\chi > 0$ (and $\chi > 0$), polarization is **left-handed**.
- If $\tan\chi < 0$ (and $\chi < 0$), polarization is **right-handed**.
- $R = 1$ (and $\chi = \pm\pi/4$) means the polarization is **circular**.
- $a_\eta = 0$ (i.e. $\chi = 0$, $R = \infty$) means the polarization is **linear**.

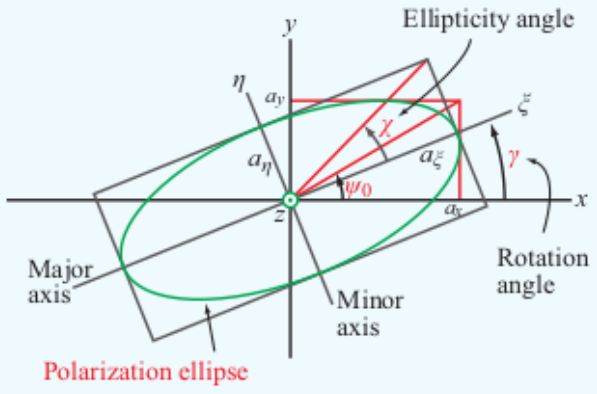


Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

Quick Recap (Continued)

- The **rotation angle** γ and **ellipticity angle** χ can be determined using a_x, a_y, and δ using

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta \tag{19}$$

$$\sin 2\chi = (\sin 2\psi_0)\sin\delta, \tag{20}$$

where

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) \in \left[0, \frac{\pi}{2}\right] \tag{21}$$

is the **auxiliary angle** and γ is required to have the same sign as $\cos\delta$. (Note: Knowing γ and χ doesn't completely specify the ellipse. Either the semi-major or semi-minor axis must also be specified.)

- I worked through an example of finding the polarization state of an elliptically polarized uniform plane wave.

Plane Wave Propagation in Lossy Media

The phasor of a plane wave propagating in a lossy medium satisfies the wave equation

$$\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = \mathbf{0}, \tag{22}$$

where

$$\gamma^2 = -\omega^2 \mu \epsilon_c \quad \text{and} \quad \epsilon_c = \epsilon' - j\epsilon'' = \epsilon - j\frac{\sigma}{\omega}. \tag{23}$$

Let α and β be the real and imaginary parts of γ so that

$$\gamma = \alpha + j\beta. \tag{24}$$

The real part α is called the **attenuation constant** and the imaginary part β is called the **phase constant**.

From the definition of γ,

$$-\omega^2 \mu \epsilon' + j(\omega^2 \mu \epsilon'') = \gamma^2 = (\alpha + j\beta)^2 = (\alpha^2 - \beta^2) + j2\alpha\beta. \tag{25}$$

Plane Wave Propagation in Lossy Media (Continued)

Therefore,

$$\alpha^2 - \beta^2 = -\omega^2 \mu \epsilon' \tag{26}$$

$$2\alpha\beta = \omega^2 \mu \epsilon''. \tag{27}$$

The second equation can be used to eliminate α or β from the first equation and solve for the remaining variable. Then that variable can be substituted into the second equation to find the other variable. The end result is

$$\alpha = \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right\}^{1/2} \text{(Np/m)} \tag{28}$$

$$\beta = \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right\}^{1/2} \text{(rad/m)} \tag{29}$$

(Note: Np = nepers, which is a logarithmic scale for ratios, like dB.)

Plane Wave Propagation in Lossy Media (Continued)

As before, assume that the phasor of the electric field intensity only has a component in the $\hat{\mathbf{x}}$ direction, i.e.

$$\tilde{\mathbf{E}}(z) = \tilde{E}_x(z)\hat{\mathbf{x}}. \tag{30}$$

The solutions to the wave equation

$$\nabla^2 \tilde{E}_x - \gamma^2 \tilde{E}_x = 0, \tag{31}$$

have exactly the same form as before, and, assuming that there is only a wave traveling in the $+z$ direction,

$$\tilde{\mathbf{E}}(z) = E_{x0} e^{-\gamma z}\hat{\mathbf{x}}. \tag{32}$$

The only difference is that γ is complex-valued whereas k was real-valued and, in this case,

$$\tilde{\mathbf{E}}(z) = E_{x0} e^{-(\alpha + j\beta)z}\hat{\mathbf{x}} = E_{x0} e^{-\alpha z} e^{-j\beta z}\hat{\mathbf{x}}. \tag{33}$$

Plane Wave Propagation in Lossy Media (Continued)

The phasor of the magnetic field intensity is

$$\tilde{\mathbf{H}}(z) = \frac{1}{\eta_c}(\hat{\mathbf{z}} \times \tilde{\mathbf{E}}) = \frac{1}{\eta_c}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & 1 \\ E_{x0}e^{-\gamma z} & 0 & 0 \end{vmatrix} = \frac{1}{\eta_c}E_{x0}e^{-\gamma z}\hat{\mathbf{y}}$$

$$= \frac{E_{x0}}{\eta_c}e^{-\alpha z}e^{-j\beta z}\hat{\mathbf{y}} = \tilde{H}_y(z)\hat{\mathbf{y}}. \tag{34}$$

Here, η_c is the **intrinsic impedance for the lossy medium**, which can be written as

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon' - j\epsilon''}} = \sqrt{\frac{\mu}{\epsilon'\left(1 - j\epsilon''/\epsilon'\right)}} = \sqrt{\frac{\mu}{\epsilon'}}\left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{-1/2}. \tag{35}$$

Since η_c is complex, $\tilde{\mathbf{E}}(z)$ and $\tilde{\mathbf{H}}(z)$ **do not have the same phase**, which means that the phases of $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ are also different.

Plane Wave Propagation in Lossy Media (Continued)

As the wave travels, its **magnitude decreases exponentially** since

$$|\tilde{E}_x(z)| = \left|E_{x0}e^{-\alpha z}e^{-j\beta z}\right| = |E_{x0}|\left|e^{-\alpha z}\right|\left|e^{-j\beta z}\right| = |E_{x0}|e^{-\alpha z} \tag{36}$$

$$|\tilde{H}_y(z)| = \left|\frac{E_{x0}}{\eta_c}e^{-\alpha z}e^{-j\beta z}\right| = \left|\frac{E_{x0}}{\eta_c}\right|\left|e^{-\alpha z}\right|\left|e^{-j\beta z}\right| = \left|\frac{E_{x0}}{\eta_c}\right|e^{-\alpha z}. \tag{37}$$

The rate of decrease α is the same for both the electric and magnetic fields and for every multiple of the **skin depth**

$$\delta_s = \frac{1}{\alpha}\ \text{m} \tag{38}$$

the magnitudes decrease by a factor of $\frac{1}{e}$. The energy lost is dissipated as heat.

Examples

- Perfect dielectric: $\sigma = 0 \implies \epsilon'' = 0 \implies \alpha = 0 \implies \delta_s = \infty$
  This means that the wave can travel infinitely far without attenuation.

Plane Wave Propagation in Lossy Media (Continued)

- Perfect conductor: $\sigma = \infty \implies \epsilon'' = \infty \implies \alpha = \infty \implies \delta_s = 0$
  This means that the magnitude is attenuated to 0 at the surface of the material and cannot penetrate into the material.

Practical application: The outer conducting material on a coaxial cable is typically several skin depths thick to prevent energy leakage from inside the cable and to prevent interference from waves outside the cable.

The expressions for α, β, and η_c are correct for any linear, isotropic, homogeneous medium.

Media are characterized by the ratio ϵ''/ϵ':

- $\epsilon''/\epsilon' \ll 1$ – **low-loss dielectric**
- $\epsilon''/\epsilon' \gg 1$ – **good conductor**
- $10^{-2} \leq \epsilon''/\epsilon' \leq 10^{2}$ – **quasi-conductor**

Plane Wave Propagation in Lossy Media (Continued)

Low-Loss Dielectrics

For this type of medium, $\epsilon''/\epsilon' \ll 1$. The propagation constant

$$\gamma = \sqrt{-\omega^2 \mu \epsilon_c} = j\omega\sqrt{\mu\left(\epsilon' - j\epsilon''\right)} = j\omega\sqrt{\mu\epsilon'\left(1 - j\frac{\epsilon''}{\epsilon'}\right)}$$
$$= j\omega\sqrt{\mu\epsilon'}\left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{1/2}. \tag{39}$$

When $|x| \ll 1$ for a complex number x, a good approximation to $\sqrt{1-x}$ is the first two terms of its Taylor series $1 - x/2$. This means

$$\left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{1/2} \approx 1 - j\frac{\epsilon''}{2\epsilon'} \tag{40}$$

Plane Wave Propagation in Lossy Media (Continued)

and

$$\gamma \approx j\omega\sqrt{\mu\epsilon'}\left(1 - j\frac{\epsilon''}{2\epsilon'}\right) = \omega\sqrt{\mu\epsilon'}\left(\frac{\epsilon''}{2\epsilon'}\right) + j\omega\sqrt{\mu\epsilon'}$$
$$= \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} + j\omega\sqrt{\mu\epsilon'}. \tag{41}$$

Recall that $\epsilon'' = \sigma/\omega$ and $\epsilon' = \epsilon$. So,

$$\gamma \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}} + j\omega\sqrt{\mu\epsilon}. \tag{42}$$

Therefore,

$$\alpha \approx \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}} \ \ (\text{Np/m}), \tag{43}$$
$$\beta \approx \omega\sqrt{\mu\epsilon'} = \omega\sqrt{\mu\epsilon} \ \ (\text{rad/m}). \tag{44}$$

Here, $\beta = k$, the wavenumber for a lossless medium.

Plane Wave Propagation in Lossy Media (Continued)

The intrinsic impedance is

$$\eta_c = \sqrt{\frac{\mu}{\epsilon'}}\left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{-1/2} \quad (\Omega). \tag{45}$$

When $|x| \ll 1$ for a complex number x, a good approximation to $(1-x)^{-1/2}$ is the first two terms of its Taylor series $1 + x/2$. Using this approximation,

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}}\left(1 + j\frac{\epsilon''}{2\epsilon'}\right) = \sqrt{\frac{\mu}{\epsilon}}\left(1 + j\frac{\sigma}{2\omega\epsilon}\right) \quad (\Omega). \tag{46}$$

Typically, $\sigma/\omega\epsilon < 10^{-2}$, which means

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega). \tag{47}$$

This is the intrinsic impedance of a lossless medium.

Plane Wave Propagation in Lossy Media (Continued)

In summary, for a **low-loss dielectric**,

- the **phase constant** β is the same as the **wavenumber** k **for a lossless medium**,
- the **intrinsic impedance** η_c is the same as the **intrinsic impedance of a lossless medium**,
- and the **attenuation constant**

$$\alpha \approx \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon'}} \quad \text{(Np/m)}. \tag{48}$$

Plane Wave Propagation in Lossy Media (Continued)

Good Conductors

In this case, $\epsilon''/\epsilon' > 100$ and

$$\alpha \approx \omega\sqrt{\frac{\mu\epsilon''}{2}} = \omega\sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\pi f \mu\sigma} \ \ (\text{Np/m}), \tag{49}$$

$$\beta \approx \alpha \approx \sqrt{\pi f \mu\sigma} \ \ (\text{rad/m}), \tag{50}$$

$$\eta_c \approx \sqrt{j\frac{\mu}{\epsilon''}} = (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma} \ \ (\Omega). \tag{51}$$

For a perfect conductor, $\sigma = \infty \implies \alpha = \beta = \infty$ and $\eta_c = 0$.

Example: Plane wave in seawater

A uniform plane wave travels in the $+z$ direction in seawater, which has parameter values

$$\epsilon_r = 80, \mu_r = 1, \text{ and } \sigma = 4 \text{ S/m}. \tag{52}$$

Plane Wave Propagation in Lossy Media (Continued)

The x-y plane is just below the surface and the magnetic field at $z = 0$ is

$$\mathbf{H}(0, t) = 100 \cos\left(\left(2\pi \times 10^3\right) t + \frac{\pi}{12}\right) \hat{\mathbf{y}} \text{ (mA/m)}. \tag{53}$$

1. Find $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$.
2. Find the depth where the amplitude of $\mathbf{E}$ is 1% of its value at $z = 0$.

1. The first step is to determine what type of medium seawater is. This means finding

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\epsilon\omega} \tag{54}$$

From the expression for $\mathbf{H}(0, t)$,

$$\omega = 2\pi \times 10^3 \text{ rad/s} \tag{55}$$

Plane Wave Propagation in Lossy Media (Continued)

so that

$$\frac{\epsilon''}{\epsilon'} \approx \frac{4 \text{ S/m}}{(80\,(1/36\pi \times 10^{-9} \text{ F/m}))\,(2\pi \times 10^{3} \text{ rad/s})}$$
$$= \frac{4 \cdot 9}{40} \times 10^{6} \text{ Ss/F} = 9 \times 10^{5}. \tag{56}$$

Since this is > 100, this means that seawater is a good conductor at the frequency $f = \omega/2\pi = 1$ kHz.

Under the given coordinate system and with $\mathbf{H}$ in the $\hat{\mathbf{y}}$ direction, the phasors of the electric and magnetic fields have the forms we have been discussing

$$\tilde{\mathbf{E}}(z) = E_{x0} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{x}} \tag{57}$$

$$\tilde{\mathbf{H}}(z) = \frac{E_{x0}}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{y}}. \tag{58}$$

So, we need to find the attenuation and phase constants and the intrinsic impedance.

Plane Wave Propagation in Lossy Media (Continued)

Using the approximations appropriate for a good conductor,

$$\alpha \approx \sqrt{\pi f \mu \sigma} = \sqrt{(\pi \times 10^3)(4\pi \times 10^{-7})(4)}$$
$$= 4\pi \times 10^{-2} \text{ Np/m} \approx 0.126 \text{ Np/m}, \tag{59}$$
$$\beta \approx \alpha \approx 4\pi \times 10^{-2} \text{ rad/m} \approx 0.126 \text{ rad/m}, \tag{60}$$

and

$$\eta_c \approx (1+j)\frac{\alpha}{\sigma} = (1+j)\frac{0.126}{4}\ \Omega. \tag{61}$$

Since

$$e^{j\pi/4} = \cos\left(\frac{\pi}{4}\right) + j\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}, \tag{62}$$
$$1 + j = \sqrt{2}e^{j\pi/4} \tag{63}$$

and

$$\eta_c \approx \sqrt{2}e^{j\pi/4}\frac{0.126}{4}\ \Omega = 0.044e^{j\pi/4}\ \Omega. \tag{64}$$

Plane Wave Propagation in Lossy Media (Continued)

The amplitude E_{x0} is complex, which means

$$E_{x0} = |E_{x0}|e^{j\phi_0}. \tag{65}$$

The electric and magnetic fields are

$$\begin{aligned}
\mathbf{E}(z,t) &= \mathrm{Re}\left\{\tilde{\mathbf{E}}(z)e^{j\omega t}\right\} = \mathrm{Re}\left\{|E_{x0}|e^{j\phi_0}e^{-\alpha z}e^{-j\beta z}e^{j\omega t}\hat{\mathbf{x}}\right\} \\
&= \mathrm{Re}\left\{|E_{x0}|e^{-\alpha z}e^{j(\omega t-\beta z+\phi_0)}\hat{\mathbf{x}}\right\} \\
&= |E_{x0}|e^{-0.126z}\cos\left(\left(2\pi\times 10^3\right)t - 0.126z + \phi_0\right)\hat{\mathbf{x}}
\end{aligned} \tag{66}$$

and

$$\begin{aligned}
\mathbf{H}(z,t) &= \mathrm{Re}\left\{\tilde{\mathbf{H}}(z)e^{j\omega t}\right\} = \mathrm{Re}\left\{\frac{|E_{x0}|e^{j\phi_0}}{\eta_c}e^{-\alpha z}e^{-j\beta z}e^{j\omega t}\hat{\mathbf{y}}\right\} \\
&= \mathrm{Re}\left\{|E_{x0}|\frac{1}{0.044e^{j\pi/4}}e^{-\alpha z}e^{j(\omega t-\beta z+\phi_0)}\hat{\mathbf{y}}\right\} \\
&= 22.5|E_{x0}|e^{-\alpha z}\mathrm{Re}\left\{e^{j(\omega t-\beta z+\phi_0-\pi/4)}\right\}\hat{\mathbf{y}}.
\end{aligned} \tag{67}$$

Plane Wave Propagation in Lossy Media (Continued)

Continuing the calculation,

$$\mathbf{H}(z,t) = 22.5|E_{x0}|e^{-0.126z}\cos\left(\left(2\pi\times10^3\right)t - 0.126z + \phi_0 - \pi/4\right)\hat{\mathbf{y}}\ \text{A/m}. \tag{68}$$

Therefore, at $z = 0$,

$$\mathbf{H}(0,t) = 22.5|E_{x0}|\cos\left(\left(2\pi\times10^3\right)t + \phi_0 - \pi/4\right)\hat{\mathbf{y}}\ \text{A/m}. \tag{69}$$

Comparing this expression to the one given in the problem

$$\mathbf{H}(0,t) = 100\cos\left(\left(2\pi\times10^3\right)t + \frac{\pi}{12}\right)\ \text{mA/m}, \tag{70}$$

$$22.5|E_{x0}|\ \text{A/m} = 100\times10^{-3}\ \text{A/m} \implies |E_{x0}| = \frac{10^{-1}}{22.5}\ \text{V/m} = 4.44\ \text{mV/m} \tag{71}$$

$$\phi_0 - \frac{\pi}{4} = \frac{\pi}{12} \implies \phi_0 = \frac{\pi}{3}. \tag{72}$$

Plane Wave Propagation in Lossy Media (Continued)

So, finally,

$$\mathbf{E}(z,t) = 4.44e^{-0.126z}\cos\left(\left(2\pi\times 10^3\right)t - 0.126z + \frac{\pi}{3}\right)\hat{\mathbf{x}}\ \text{mV/m} \quad (73)$$

$$\mathbf{H}(z,t) = 100e^{-0.126z}\cos\left(\left(2\pi\times 10^3\right)t - 0.126z + \frac{\pi}{12}\right)\hat{\mathbf{y}}\ \text{mA/m}. \quad (74)$$

Notice that the wave is still TEM but the electric and magnetic fields are no longer in phase.

2. The depth z at which the amplitude of $\mathbf{E}(z,t)$ is 1% its value at $z = 0$ satisfies

$$4.44e^{-0.126z} = 0.01\cdot 4.44e^{-0.126z}\Big|_{z=0} = 0.01\cdot 4.44. \quad (75)$$

This means

$$e^{-0.126z} = 0.01 \quad (76)$$

and

$$z = -\frac{1}{0.126}\ln 0.01 = 36.5\ \text{m}. \quad (77)$$

Current Flow in a Good Conductor

The behavior of **uniform plane waves in a good conductor** implies that the **flow of current** in this type of material is **very different** when the voltage source is **static**, i.e. dc, **versus time-varying**, i.e. ac. For a **dc voltage source**, the **current is evenly distributed** throughout the conductor but, for an **ac voltage source**, the **current mainly flows through the surface**.

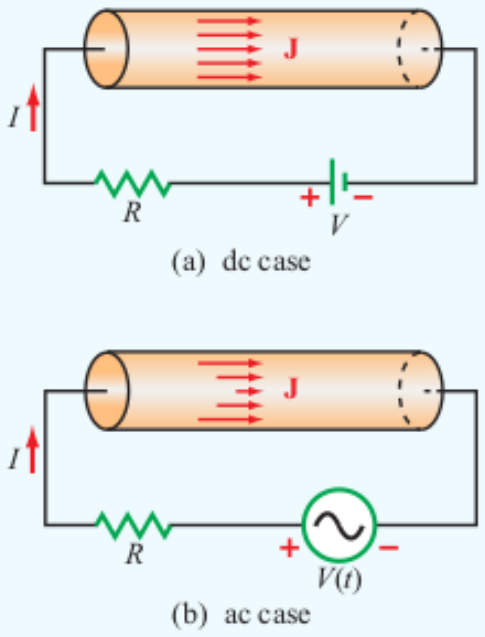


Figure 7-14 Current density **J** in a conducting wire is (a) uniform across its cross section in the dc case, but (b) in the ac case, **J** is highest along the wire's perimeter.

Current Flow in a Good Conductor (Continued)

To see why, consider a **solid, semi-infinite conductor bounded above by a perfect dielectric**, where the boundary between the materials is the x-y plane and the z axis points into the conductor. Suppose that there is an x-**polarized electric field in the dielectric material** just above the surface (at $z = 0^-$) with

$$\tilde{\mathbf{E}} = E_0 \hat{\mathbf{x}}. \tag{78}$$

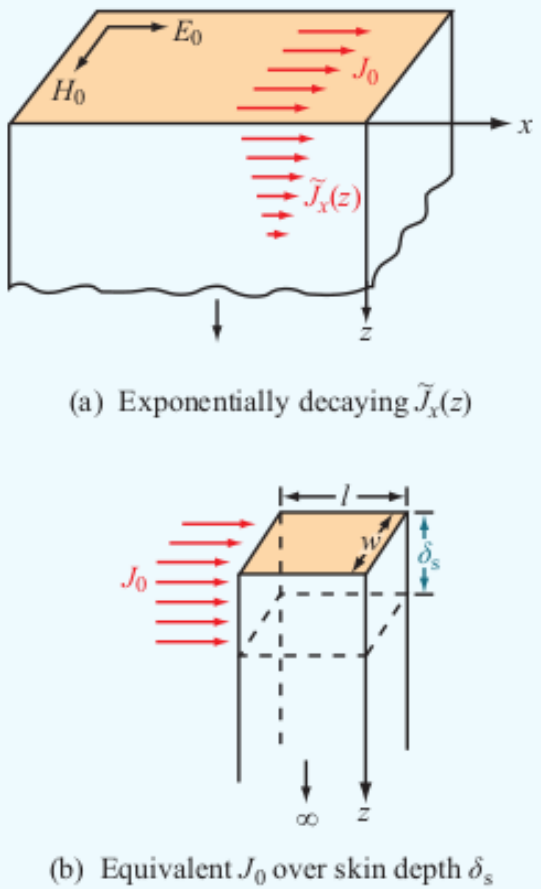


Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

Current Flow in a Good Conductor (Continued)

This **induces an electric field** with phasor $\tilde{\mathbf{E}}(z)$ **in the conductor** with

$$\tilde{\mathbf{E}}(z = 0^+) = E_0 \hat{\mathbf{x}} \tag{79}$$

since the tangential component of $\mathbf{E}$ is continuous across the boundary.

Based on the discussion last lecture,

$$\tilde{\mathbf{E}}(z) = E_0 e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{x}} \tag{80}$$

$$\tilde{\mathbf{H}}(z) = \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{y}}. \tag{81}$$

The resulting current in the conductor satisfies Ohm's law $\mathbf{J} = \sigma \mathbf{E}$

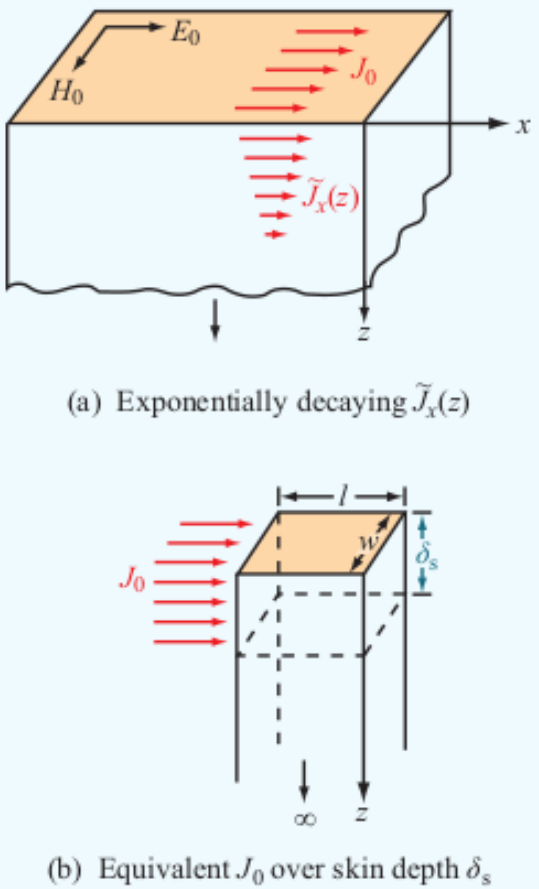


Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

Current Flow in a Good Conductor (Continued)

so that

$$\tilde{\mathbf{J}}(z) = \tilde{J}_x(z)\hat{\mathbf{x}}, \tag{82}$$

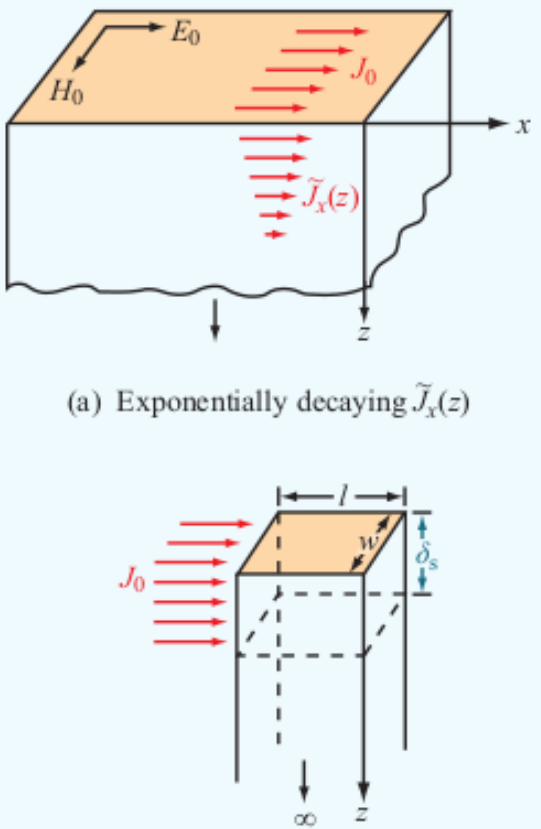
where

$$\begin{aligned}\tilde{J}_x(z) &= \sigma E_0 e^{-\alpha z} e^{-j\beta z} \\ &= J_0 e^{-\alpha z} e^{-j\beta z}.\end{aligned} \tag{83}$$

Here,

$$J_0 = \sigma E_0 \tag{84}$$

is the **current density in the conductor at the boundary**.



(a) Exponentially decaying $\tilde{J}_x(z)$

(b) Equivalent J_0 over skin depth δ_s

Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

Current Flow in a Good Conductor (Continued)

The **skin depth** is

$$\delta_s = 1/\alpha \tag{85}$$

and, **in a good conductor**,

$$\beta \approx \alpha. \tag{86}$$

So,

$$\begin{aligned}\tilde{J}_x(z) &= J_0 e^{-\alpha z} e^{-j\beta z} = J_0 e^{-(\alpha + j\beta)z} \\ &= J_0 e^{-(\alpha + j\alpha)z} = J_0 e^{-\alpha(1+j)z} \\ &= J_0 e^{-(1+j)z/\delta_s} \ (\mathrm{A/m^2}).\end{aligned} \tag{87}$$

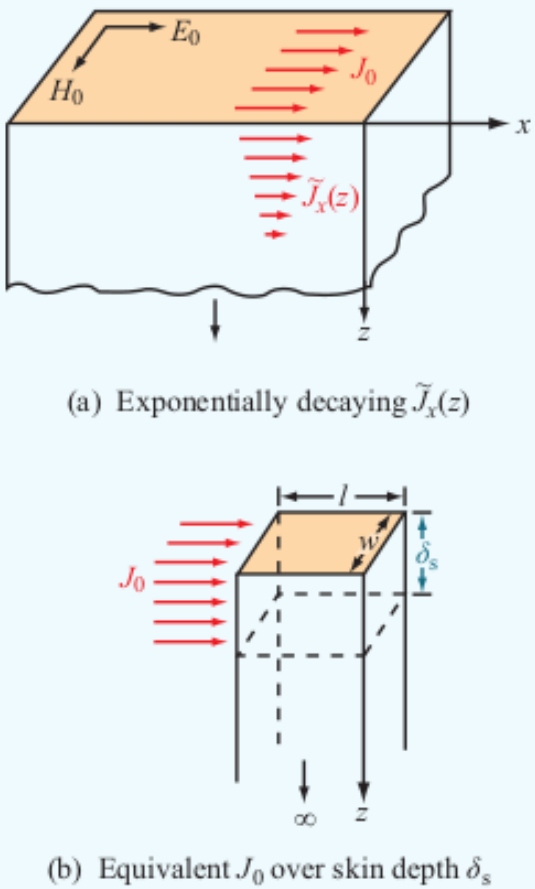


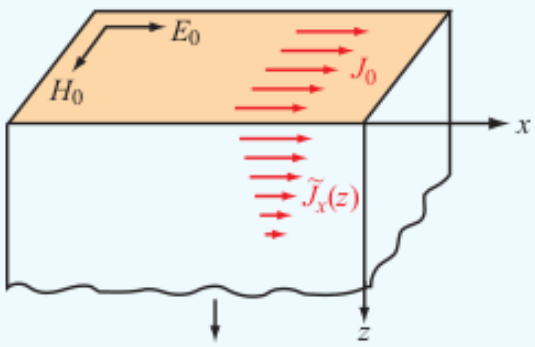
Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

Current Flow in a Good Conductor (Continued)

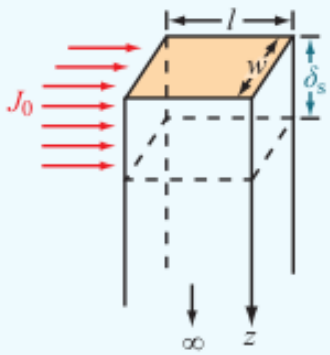
The phasor of the **total current passing through a slice with width w in the y direction** that extends the entire depth of the conductor is

$$\begin{aligned}\tilde{I} &= w\int_0^\infty \tilde{J}_x(z)\,dz \\ &= w\int_0^\infty J_0 e^{-(1+j)z/\delta_s}\,dz \\ &= J_0 w\left(-\frac{\delta_s}{1+j}e^{-(1+j)z/\delta_s}\Big|_0^\infty\right) \\ &= \frac{J_0 w\delta_s}{1+j}. \end{aligned} \tag{88}$$

If the slice has a finite thickness $>$ a few times the skin depth, say $5\delta_s$, this is approximately true with a small error.



(a) Exponentially decaying $\tilde{J}_x(z)$



(b) Equivalent J_0 over skin depth δ_s

Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

Homework

Read Chapter 7.

Homework 1 is due now.

Homework 2 is posted and will be due on Tuesday, May 5.

Exam 1 will be Thursday, May 7.